translationBridge MQTT/TLS/JSON

Interface Design Description

Service ID: *"translationBridge"*

Abstract
This document describes the MQTT protocol with TLS payload security and JSON payload encoding variant of the `translationBridge` service of the TranslationManager system.

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `translationBridge` service interface of the TranslationManager system over MQTT, which provides interface and data-model translation for a consumer: finding providers that become usable through a translation bridge, building the bridge and aborting it.

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	MQTT	3.1.1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `translationBridge` service interface

This document provides the Interface Design Description IDD to the *translationBridge – Service Description* document [5]. The operations, their semantics and the data models are the same as in the HTTP/TLS/JSON IDD [6]; this document specifies the MQTT binding.

The MQTT interface is offered only when the system runs with `mqtt.api.enabled=true`; it is then registered with the interface template `generic_mqtts` (`generic_mqtt` without TLS), the broker address and port (`mqtt.broker.address`, `mqtt.broker.port`) and the base topic `arrowhead/translation/bridge/`. The standalone deployment sets `mqtt.api.enabled=false`, so the MQTT interfaces are not registered there.

3 Service Operations

The provider subscribes to one topic per operation, the base topic followed by the operation name:

Operation	Request topic
`discovery`	`arrowhead/translation/bridge/discovery`
`negotiation`	`arrowhead/translation/bridge/negotiation`
`abort`	`arrowhead/translation/bridge/abort`

A consumer publishes a request envelope (MqttRequestTemplate) to the topic and receives the answer (MqttResponseTemplate) on its own `responseTopic`. The `authentication` field carries the requester's identity according to the provider's authentication policy: `IDENTITY-TOKEN//<token>` (outsourced or internal policy), `SYSTEM//<name>` (declared policy) or the requester's X.509 certificate (certificate policy). The `status` of the response uses the HTTP status code numbers (200, 201, 204, 400, 401, 403, 404, 408, 423, 500, 503); on error the payload is an ErrorMessage.

```
{
  "traceId": "a1b2c3",
  "authentication": "IDENTITY-TOKEN//<token>",
  "responseTopic": "arrowhead/response/TemperatureDisplay/a1b2c3",
  "qosRequirement": 1,
  "params": {},
  "payload": {}
}
```

Listing 1: Request envelope.

```
{
  "status": 200,
  "traceId": "a1b2c3",
  "receiver": "TemperatureDisplay",
  "payload": {}
}
```

Listing 2: Response envelope.

3.1 PUBLISH `arrowhead/translation/bridge/discovery`

Operation: discovery
Input: TranslationDiscoveryRequest
Output: TranslationDiscoveryResponse

Returns the candidates usable by the requester through translation, with a bridge identifier.

```
{
  "candidates": [
    {
      "instanceId": "TemperatureSensor|temperatureReading|1.0.0",
      "provider": {
        "name": "TemperatureSensor",
        "metadata": {
          "location": "hall-1"
        },
        "version": "1.0.0",
        "addresses": [
          {}
        ],
        "device": {
```

```
          "name": "Raspberry1",
          "metadata": {
            "location": "hall-1"
          },
          "addresses": [
            {}
          ],
          "createdAt": "2026-10-22T12:00:00Z",
          "updatedAt": "2026-10-22T12:00:00Z"
        },
        "createdAt": "2026-10-22T12:00:00Z",
        "updatedAt": "2026-10-22T12:00:00Z"
      },
      "serviceDefinition": {
        "name": "temperatureReading",
        "createdAt": "2026-10-22T12:00:00Z",
        "updatedAt": "2026-10-22T12:00:00Z"
      },
      "version": "1.0.0",
      "expiresAt": "2026-10-22T12:00:00Z",
      "metadata": {
        "unit": "celsius"
      },
      "interfaces": [
        {
          "templateName": "generic_https",
          "protocol": "https",
          "policy": "CERT_AUTH",
          "properties": {
            "accessAddresses": [
              "192.168.0.10"
            ],
            "accessPort": 8080,
            "basePath": "/temperature",
            "operations": {
              "read": {
                "method": "GET",
                "path": "/read"
              }
            }
          }
        }
      ],
      "createdAt": "2026-10-22T12:00:00Z",
      "updatedAt": "2026-10-22T12:00:00Z"
    }
  ],
  "operation": "FILTER",
  "interfaceTemplateNames": [
    "string"
  ],
  "inputDataModelId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "outputDataModelId": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
}
```


Listing 3: Payload of an discovery request.

3.2 PUBLISH `arrowhead/translation/bridge/negotiation`

Operation: negotiation
Input: TranslationNegotiationRequest
Output: TranslationNegotiationResponse

Builds the bridge to the chosen candidate and returns the bridge interface and token limits.


```
{
```

```
  "bridgeId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "target": {
    "instanceId": "TemperatureSensor|temperatureReading|1.0.0",
    "provider": {
      "name": "TemperatureSensor",
      "metadata": {
        "location": "hall-1"
      },
      "version": "1.0.0",
      "addresses": [
        {
          "type": "IPV4",
          "address": "192.168.0.10"
        }
      ],
      "device": {
        "name": "Raspberry1",
        "metadata": {
          "location": "hall-1"
        },
        "addresses": [
          {}
        ],
        "createdAt": "2026-10-22T12:00:00Z",
        "updatedAt": "2026-10-22T12:00:00Z"
      },
      "createdAt": "2026-10-22T12:00:00Z",
      "updatedAt": "2026-10-22T12:00:00Z"
    },
    "serviceDefinition": {
      "name": "temperatureReading",
      "createdAt": "2026-10-22T12:00:00Z",
      "updatedAt": "2026-10-22T12:00:00Z"
    },
    "version": "1.0.0",
    "expiresAt": "2026-10-22T12:00:00Z",
    "metadata": {
      "unit": "celsius"
    },
    "interfaces": [
      {
        "templateName": "generic_https",
        "protocol": "https",
        "policy": "CERT_AUTH",
        "properties": {
          "accessAddresses": [
            "192.168.0.10"
          ],
          "accessPort": 8080,
          "basePath": "/temperature",
          "operations": {
            "read": {
              "method": "GET",
              "path": "/read"
            }
          }
        }
      }
    ],
    "createdAt": "2026-10-22T12:00:00Z",
    "updatedAt": "2026-10-22T12:00:00Z"
  },
  "operation": "FILTER",
  "interfaceTemplateName": "string",
  "inputDataModelId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "outputDataModelId": "3fa85f64-5717-4562-b3fc-2c963f66afa6"
}
```


Listing 4: Payload of an negotiation request.

3.3 PUBLISH `arrowhead/translation/bridge/abort`

Operation: abort
Input: String

Aborts a bridge of the requester. Returns 200 if aborted, 204 if there was nothing to abort. The bridge identifier fills the `{bridgeId}` placeholder of the registered path.

```
"3fa85f64-5717-4562-b3fc-2c963f66afa6"
```

Listing 5: Payload of an abort request.

4 Data Models

Here, all data objects that can be part of the service calls associated with this service are listed in alphabetic order. Each *struct* denotes a JSON Object with the named fields; fields that are not needed may be omitted.

4.1 struct MqttRequestTemplate

The request envelope.

Object Field	Value Type	Description
"traceId"	String	Identifier chosen by the requester, returned in the response.
"authentication"	String	Identity of the requester (see Section 3).
"responseTopic"	String	Topic on which the response is published.
"qosRequirement"	Number	MQTT QoS of the response (0, 1 or 2).
"params"	Object<String>	Optional parameters (the query parameters of the HTTP binding).
"payload"	Object	The operation input.

4.2 struct MqttResponseTemplate

The response envelope.

Object Field	Value Type	Description
"status"	Number	Status code.
"traceId"	String	The trace identifier of the request.
"receiver"	String	Name of the requester.
"payload"	Object	The operation output, or an ErrorMessage.

4.3 struct ErrorMessage

Object Field	Value Type	Description
"errorMessage"	String	Error text, if the event is an error.
"errorCode"	Number	
"exceptionType"	ExceptionType	
"origin"	String	

4.4 struct Address

A network address with its type.

Object Field	Value Type	Description
"type"	String	Address type: `HOSTNAME`, `IPV4`, `IPV6` or `MAC`.
"address"	String	One network address (host name, IPv4, IPv6 or MAC address).

4.5 struct DeviceResponse

A registered device.

Object Field	Value Type	Description
"name"	String	Device name.
"metadata"	Object	Free-form metadata (JSON object).
"addresses"	Array<Address>	Network addresses with their types.
"createdAt"	DateTime	UTC time of creation.
"updatedAt"	DateTime	UTC time of the last change.

4.6 struct ServiceDefinitionResponse

A service definition.

Object Field	Value Type	Description
"name"	String	Service definition name.
"createdAt"	DateTime	UTC time of creation.
"updatedAt"	DateTime	UTC time of the last change.

4.7 struct ServiceInstanceInterfaceResponse

One interface of a service instance.

Object Field	Value Type	Description
"templateName"	String	Name of the interface template, snake_case (e.g. `generic_https`).
"protocol"	String	Protocol of the interface, e.g. `http`, `https`, `mqtt`, `mqtts`.
"policy"	String	Security policy of the interface: `NONE`, `CERT_AUTH` or one of the token policies.
"properties"	Object	Interface properties defined by the template, e.g. `accessAddresses`, `accessPort`, `basePath`, `operations`.

4.8 struct ServiceInstanceResponse

A registered service instance.

Object Field	Value Type	Description
"instanceId"	String	Service instance identifier `<System>|<serviceDefinition>|<version>`, or policy instance identifier.
"provider"	SystemResponse	The provider system.
"serviceDefinition"	ServiceDefinitionResponse	The service definition.
"version"	Version	Version of the service instance.
"expiresAt"	DateTime	UTC expiry time; no value means no expiry.
"metadata"	Object	Free-form metadata (JSON object).
"interfaces"	Array<ServiceInstanceInterface Response>	Service interfaces offered by the instance.
"createdAt"	DateTime	UTC time of creation.
"updatedAt"	DateTime	UTC time of the last change.

4.9 struct SystemResponse

A registered system.

Object Field	Value Type	Description
"name"	String	System name.
"metadata"	Object	Free-form metadata (JSON object).
"version"	Version	Version of the system.
"addresses"	Array<Address>	Network addresses with their types.
"device"	DeviceResponse	The device hosting the system, if registered.
"createdAt"	DateTime	UTC time of creation.
"updatedAt"	DateTime	UTC time of the last change.

4.10 struct TranslationBridgeCandidate

A candidate reachable through translation.

Object Field	Value Type	Description
"serviceInstanceId"	String	Service instance identifier `<System>|<serviceDefinition>|<version>`.
"interfaceTemplate Name"	String	Interface template of the candidate the bridge would translate to.

4.11 struct TranslationDiscoveryRequest

Discovery of translation bridges for the requester.

Object Field	Value Type	Description
"candidates"	Array<ServiceInstanceResponse>	Candidate service instances (from service discovery).
"operation"	String	Operation the consumer wants to call.
"interfaceTemplate Names"	Array<String>	Interface templates the consumer can use.
"inputDataModelId"	String	Data model of the operation input expected by the consumer.
"outputDataModelId"	String	Data model of the operation output expected by the consumer.

4.12 struct TranslationDiscoveryResponse

Candidates reachable through a translation bridge.

Object Field	Value Type	Description
"bridgeId"	String	Identifier (UUID) of the translation bridge.
"candidates"	Array<TranslationBridgeCandidate>	Store entries to create.

4.13 struct TranslationNegotiationRequest

Builds the bridge to the chosen candidate.

Object Field	Value Type	Description
"bridgeId"	String	Identifier (UUID) of the translation bridge.
"target"	ServiceInstanceResponse	The chosen candidate service instance.
"operation"	String	QoS operation: `FILTER` or `SORT`.
"interfaceTemplate Name"	String	Interface template the consumer will use.
"inputDataModelId"	String	Data model of the operation input expected by the consumer.
"outputDataModelId"	String	Data model of the operation output expected by the consumer.

4.14 struct TranslationNegotiationResponse

The established bridge.

Object Field	Value Type	Description
"bridgeId"	String	Identifier (UUID) of the translation bridge.
"bridgeInterface"	ServiceInstanceInterfaceResponse	Interface of the bridge the consumer calls instead of the provider.
"tokenExpiresAt"	DateTime	Expiry of the bridge token.
"tokenUsageLimit"	Number	Number of uses allowed with the bridge token.

4.15 Primitives

As all messages are encoded using the JSON format [4], the following primitive constructs, part of that standard, become available.

Type	Description
Array	A JSON Array [4]; Array<A> has elements of type A.
DateTime	JSON String with an ISO 8601 UTC timestamp, e.g. `2026-10-22T12:00:00Z`.
ExceptionType	One of `ARROWHEAD`, `INVALID_PARAMETER`, `AUTH`, `FORBIDDEN`, `DATA_NOT_FOUND`, `TIMEOUT`, `LOCKED`, `INTERNAL_SERVER_ERROR`, `EXTERNAL_SERVER_ERROR` (JSON String).
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.
Version	JSON String, semantic version `major.minor.patch`.

5 Go and Rust implementation variants

5.1 Go implementation

Primary system folder: `Eclipse-Arrowhead-go/ah5-TranslationManager-go`.

The system owns its Go module, launcher and build configuration. Use the per-system `Makefile` and `go.mod`; the containing Go tree is the build source, not a historical Arrowhead copy. Ordinary native roles use independent identities and JSON snapshot state rather than the Java database. Do not assume Java SQL migration, clustering or identical transaction/retry semantics. The registry has its own configuration and optional native AMQP monitor; other roles must be checked against their own registrations.

Listening port. Built-in default 127.0.0.1:8465 (Arrowhead default). The shipped config.json, compose.yaml and Dockerfile use the same port on all interfaces ("listen": "0.0.0.0:8465"; 8540–8555 layout before 2026-11-09). Override with "listen" in the configuration file, SERVER_PORT=N or -server.port N; the port overrides keep the configured host.

Native behavior and outstanding work. Port of the Java TranslatorBridgeEngine (2026-10-24): registry discovery of interface translators (interfaceBridgeManagement), data model translators (dataModelTranslation) and factories (dataModelTranslatorFactoryControl); check-targets / initialize-bridge / abort-bridge; Java bridge statuses, transitions, reports, query and 8-hour cleanup; consumer and management APIs. Translators with TOKEN_AUTH interfaceBridgeManagement policies are not selected; custom translator configuration (TODO in Java) not sent; first-match matchmaking. These are the boundaries recorded by the native compatibility document; later implementation changes still require direct qualification. Native design/route evidence: `Eclipse-Arrowhead-go/ah5-TranslationManager-go/documentation/TranslationManager`.

For this Java interface, select only a native transport and operation explicitly implemented and advertised by that port. A matching method/path does not establish DTO, error, authorization, retry or persistence equivalence. Java MQTT/AMQP bindings are not automatically provided by a Go or Rust role. Differential contract and security checks remain necessary before substitution.

5.2 Rust implementation

Primary system folder: `Eclipse-Arrowhead-rust/ah5-TranslationManager-rust`.

The per-system entry point is declared as a binary of `arrowhead-support-rust` in `ah5-support-runtime-rust/Cargo.toml`. Build from the primary Rust workspace with the applicable package/binary selection. Documentation links can resolve into the shared core/support catalogue. Independent process identity and state remain required.

The ordinary native profile uses Tokio/Axum and rustls with peer-identity checks and independent locked JSON state. Network effects are not distributed database transactions. The precise trust, CRL reload and persistence guarantees are those of the native profile, not the Java implementation.

Listening port. Default 127.0.0.1:8465 (Arrowhead default, role translationmanager). Override with "listen": "host:port" in the configuration, SERVER_PORT=N or –server.port N; the port overrides keep the configured host.

Native behavior and outstanding work. Durable bridge lifecycle through pinned HTTPS InterfaceTranslator adapter. Java translator chaining/native conversion/discovery; absent adapter returns503; uncertain effects require reconciliation. These are the boundaries recorded by the native compatibility document; later implementation changes still require direct qualification. Native design/route evidence: `Eclipse-Arrowhead-rust/ah5-support-runtime-rust/documentation/TranslationManager`.

For this Java interface, select only a native transport and operation explicitly implemented and advertised by that port. A matching method/path does not establish DTO, error, authorization, retry or persistence equivalence. Java MQTT/AMQP bindings are not automatically provided by a Go or Rust role. Differential contract and security checks remain necessary before substitution.

6 References

[1] A. Banks and R. Gupta, “Mqtt version 3.1.1,” OASIS, Standard, 2014.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “translationBridge – Service Description (TranslationManager),” Eclipse Arrowhead, SD 5.3.0, 2026, translationManager-v5.3.0-translationBridge-SD.pdf.

[6] ——, “translationBridge HTTP/TLS/JSON – Interface Design Description (TranslationManager),” Eclipse Arrowhead, IDD 5.3.0, 2026, translationManager-v5.3.0-translationBridge-HTTP-IDD.pdf.

7 Revision History

7.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): standalone deployment, configuration and security statements updated; missing primitive definitions added.	Claude Opus 5.5 (draft)
3	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

7.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	